

MATH3303 – Lecture 24

Ring Isomorphisms and the Isomorphism Theorems

Review Notes

Reference: Lee, Abstract Algebra, Sections 9.4–9.5

Overview

This lecture continues from Lecture 23's framework of ring homomorphisms, kernels, and ideals, and develops the three isomorphism theorems for rings:

- **More examples of kernels** – the reduction map $\mathbb{Z} \rightarrow \mathbb{Z}_n$ and the evaluation map $\mathbb{Z}[x] \rightarrow \mathbb{Z}$
- **Ring isomorphism** – a bijective ring homomorphism; canonical example $\mathbb{C} \hookrightarrow M_2(\mathbb{R})$
- **First Isomorphism Theorem** – $R/\ker \varphi \cong \operatorname{Im} \varphi$, with applications to $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$, $K[x]/(x-a) \cong K$, $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$, and the Chinese Remainder Theorem
- **Second Isomorphism Theorem** – $(I+J)/J \cong I/(I \cap J)$
- **Third Isomorphism Theorem** – $(R/I)/(J/I) \cong R/J$ when $I \subseteq J$

Connection to Lecture 23. Lecture 23 introduced ring homomorphisms and established that $\ker \varphi$ is a two-sided ideal of R while $\operatorname{Im} \varphi$ is a subring of S . The First Isomorphism Theorem packages these facts into a tight structural statement: every ring homomorphism factors through its kernel, and the resulting quotient is isomorphic to the image. The Second and Third theorems then describe how ideals and quotients interact.

1 More Examples of Ring Homomorphisms

We begin with two examples that illustrate how to compute kernels, both of which will reappear in the proof of the First Isomorphism Theorem.

Example 1.1: Reduction modulo n (from lecture; cf. Lee, Example 9.12)

Define $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ by $\varphi(a) = \bar{a}$ (the residue class of a modulo n). This is a surjective ring homomorphism, and

$$\ker \varphi = n\mathbb{Z}.$$

Example 1.2: Evaluation at 0 on $\mathbb{Z}[x]$ (from lecture)

Define $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}$ by $\varphi(f(x)) = f(0)$. This is a ring homomorphism. Its kernel is the ideal of polynomials with zero constant term:

$$\ker \varphi = (x).$$

2 Ring Isomorphisms

Definition 2.1: Ring isomorphism (from lecture; cf. Lee, Def. 9.6)

Let R and S be rings. A *ring isomorphism* is a bijective ring homomorphism $f : R \rightarrow S$. When such an f exists we write $R \cong S$ and say that R and S are *isomorphic* as rings.

Example 2.1: $\mathbb{C}$ embeds into $M_2(\mathbb{R})$ (from lecture; cf. Lee, Example 9.18)

Define

$$f : \mathbb{C} \rightarrow S = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : a, b \in \mathbb{R} \right\}, \quad f(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

We claim f is a ring isomorphism between $\mathbb{C}$ and S .

Check. f is injective and surjective by construction (the entries a, b are read off uniquely from the matrix). It remains to verify the homomorphism axioms.

Addition. For $z = a + bi$ and $w = c + di$ in $\mathbb{C}$,

$$f(z + w) = f((a + c) + (b + d)i) = \begin{pmatrix} a + c & b + d \\ -(b + d) & a + c \end{pmatrix} = f(z) + f(w).$$

Multiplication.

$$\begin{aligned} f((a + bi)(c + di)) &= f((ac - bd) + (ad + bc)i) \\ &= \begin{pmatrix} ac - bd & ad + bc \\ -(ad + bc) & ac - bd \end{pmatrix} \\ &= \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \begin{pmatrix} c & d \\ -d & c \end{pmatrix} = f(a + bi) f(c + di). \end{aligned}$$

Identity. $f(1) = f(1 + 0 \cdot i) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, the identity of S .

So $\mathbb{C} \cong S$ as rings.

3 The First Isomorphism Theorem

Theorem 3.1: First Isomorphism Theorem for rings (from lecture; cf. Lee, Thm. 9.17)

Let R and S be rings and $\varphi : R \rightarrow S$ a ring homomorphism. Then

$$R/\ker \varphi \cong \operatorname{Im} \varphi.$$

Proof. Set $K = \ker \varphi$ (a two-sided ideal of R , by Lecture 23). Define

$$\bar{\varphi} : R/K \rightarrow \operatorname{Im} \varphi, \quad \bar{\varphi}(r + K) = \varphi(r).$$

Step 1: $\bar{\varphi}$ is an isomorphism of additive groups. This follows from the First Isomorphism Theorem for groups applied to φ as a homomorphism of the underlying additive groups; in particular, well-definedness, bijectivity, and additivity are all immediate.

Step 2: $\bar{\varphi}$ respects multiplication. For $r, s \in R$,

$$\bar{\varphi}((r + K)(s + K)) = \bar{\varphi}(rs + K) = \varphi(rs) = \varphi(r)\varphi(s) = \bar{\varphi}(r + K)\bar{\varphi}(s + K).$$

Step 3: $\bar{\varphi}$ sends identity to identity.

$$\bar{\varphi}(1 + K) = \varphi(1),$$

which is the multiplicative identity of $\operatorname{Im} \varphi$ (by the proposition in Lecture 23, or directly because $\varphi(1)$ is the identity of S under our lecturer's convention).

Hence $\bar{\varphi}$ is a ring isomorphism, and $R/K \cong \operatorname{Im} \varphi$. □

Applications of the First Isomorphism Theorem

Example 3.1: $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$ (from lecture; cf. Lee, Example 9.24)

Recall $\mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$ is the ring with addition and multiplication modulo n . Define

$$\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_n, \quad a \mapsto \bar{a}.$$

This is a ring homomorphism. It is surjective, and $\ker \varphi = (n) = n\mathbb{Z}$. By the First Isomorphism Theorem,

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n.$$

Example 3.2: Evaluation homomorphism: $K[x]/(x - a) \cong K$ (from lecture; cf. Lee, Example 9.25)

Let K be a field and fix $a \in K$. Define the *evaluation homomorphism*

$$\varphi_a : K[x] \rightarrow K, \quad \varphi_a(f) = f(a).$$

This is a ring homomorphism (a special case of the evaluation lemma, cf. Lee, Lemma 11.1). It is surjective because every constant polynomial is mapped to itself.

Claim: $\ker \varphi_a = (x - a)$.

($\supseteq$) Clear: any multiple of $x - a$ vanishes at a .

($\subseteq$) Let $f(x) \in \ker \varphi_a$, so $f(a) = 0$. By the division algorithm in $K[x]$, there exist $q(x) \in K[x]$ and $r \in K$ (constant, because $\deg r < \deg(x - a) = 1$) such that

$$f(x) = (x - a)q(x) + r.$$

Evaluating at a gives $0 = f(a) = r$, so $r = 0$. Hence $f(x) = (x - a)q(x) \in (x - a)$.

By the First Isomorphism Theorem,

$$K[x]/(x - a) \cong K.$$

Exercise 3.1: $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$ (from lecture; cf. Lee, Example 9.25 in spirit)

Define $\varphi : \mathbb{R}[x] \rightarrow \mathbb{C}$ by $\varphi(f(x)) = f(i)$, a ring homomorphism. Show that $\ker \varphi = (x^2 + 1)$ and conclude that

$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}.$$

Example 3.3: Chinese Remainder Theorem (from lecture; cf. Lee, Thm. 2.13 and Exercise 9.42)

Let $m, n \in \mathbb{Z}$ be coprime, i.e. $\gcd(m, n) = 1$. Define the ring homomorphism

$$\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_m \times \mathbb{Z}_n, \quad a \mapsto (\bar{a} \pmod{m}, \bar{a} \pmod{n}).$$

Then

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}_m \times \mathbb{Z}_n \quad (\text{Chinese Remainder Theorem}).$$

Check (1): φ is surjective. Surjectivity is equivalent to solving the simultaneous system

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}.$$

Since $\gcd(m, n) = 1$, there exist $s, t \in \mathbb{Z}$ with

$$ms + nt = 1.$$

Set $x = ant + bms$. Then

$$x = a(1 - ms) + bms \equiv a \pmod{m},$$

$$x = ant + b(1 - nt) \equiv b \pmod{n}.$$

Hence φ is surjective.

Check (2): $\ker \varphi = mn\mathbb{Z}$.

($\supseteq$) Clearly $mn\mathbb{Z} \subseteq \ker \varphi$, since any multiple of mn is divisible by both m and n .

($\subseteq$) Let $a \in \ker \varphi$, so $\varphi(a) = (0 \pmod{m}, 0 \pmod{n})$. Then $m \mid a$ and $n \mid a$. Since $\gcd(m, n) = 1$, it follows that $mn \mid a$, i.e. $a \in mn\mathbb{Z}$. Hence $\ker \varphi \subseteq mn\mathbb{Z}$.

Therefore $\ker \varphi = mn\mathbb{Z}$, and by the First Isomorphism Theorem,

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}_m \times \mathbb{Z}_n.$$

4 The Second Isomorphism Theorem

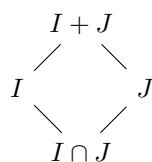
Theorem 4.1: Second Isomorphism Theorem (from lecture; cf. Lee, Thm. 9.18)

Let I, J be (two-sided) ideals of R . Then:

- (1) $I + J \trianglelefteq R$ and $I \cap J \trianglelefteq I$;
- (2) $(I + J)/J \cong I/(I \cap J)$,

where the isomorphism is induced by $a + J \mapsto a + I \cap J$ (for $a \in I$).

The lattice picture is the standard diamond:



The two “parallel sides” of the diamond correspond to isomorphic quotients.

Example 4.1: Concrete instance in $\mathbb{Z}$ (from lecture; cf. Lee, Example 9.26)

Let $R = \mathbb{Z}$, $I = 2\mathbb{Z}$, $J = 3\mathbb{Z}$. Recall the general facts:

$$m\mathbb{Z} + n\mathbb{Z} = \gcd(m, n)\mathbb{Z}, \quad m\mathbb{Z} \cap n\mathbb{Z} = \text{lcm}(m, n)\mathbb{Z}.$$

So

$$I + J = 2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}, \quad I \cap J = 2\mathbb{Z} \cap 3\mathbb{Z} = 6\mathbb{Z}.$$

The Second Isomorphism Theorem gives

$$\frac{2\mathbb{Z} + 3\mathbb{Z}}{3\mathbb{Z}} \cong \frac{2\mathbb{Z}}{6\mathbb{Z}}, \quad \text{i.e.} \quad \mathbb{Z}/3\mathbb{Z} \cong 2\mathbb{Z}/6\mathbb{Z}.$$

5 The Third Isomorphism Theorem

Theorem 5.1: Third Isomorphism Theorem (from lecture; cf. Lee, Thm. 9.19)

Let $I \subseteq J$ be ideals of R . Then:

- (1) $J/I \leq R/I$;
- (2) $(R/I)/(J/I) \cong R/J$,

where the isomorphism is given by $(r + I) + J/I \mapsto r + J$.

Example 5.1: Concrete instance in $\mathbb{Z}$ (from lecture)

Let $R = \mathbb{Z}$, $I = 6\mathbb{Z}$, $J = 2\mathbb{Z}$. Note $6\mathbb{Z} \subseteq 2\mathbb{Z}$. The Third Isomorphism Theorem gives

$$\frac{\mathbb{Z}/6\mathbb{Z}}{2\mathbb{Z}/6\mathbb{Z}} \cong \mathbb{Z}/2\mathbb{Z}.$$

Checklist of Skills

- ☐ Compute the kernel of a ring homomorphism, e.g. $\ker(\mathbb{Z} \rightarrow \mathbb{Z}_n) = n\mathbb{Z}$ and $\ker(\mathbb{Z}[x] \rightarrow \mathbb{Z}, f \mapsto f(0)) = (x)$.
- ☐ State the definition of a ring isomorphism and verify all four conditions (injective, surjective, additive, multiplicative; identity preserved).
- ☐ Reproduce the embedding $\mathbb{C} \hookrightarrow M_2(\mathbb{R})$ as the canonical example of a ring isomorphism onto its image.
- ☐ State and prove the First Isomorphism Theorem for rings.
- ☐ Apply the First Isomorphism Theorem to standard quotients: $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$, $K[x]/(x-a) \cong K$, $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$.
- ☐ Carry out the Chinese Remainder Theorem proof in the ring-theoretic framing: surjectivity via Bézout, kernel via $\gcd(m, n) = 1 \implies mn \mid a$.
- ☐ State the Second Isomorphism Theorem and draw the diamond lattice $I \subseteq I+J \supseteq J \supseteq I \cap J$.
- ☐ State the Third Isomorphism Theorem and recognise the “cancel the inner quotient” pattern $(R/I)/(J/I) \cong R/J$.
- ☐ Apply the Second and Third Theorems to concrete examples in $\mathbb{Z}$ using $m\mathbb{Z} + n\mathbb{Z} = \gcd \mathbb{Z}$ and $m\mathbb{Z} \cap n\mathbb{Z} = \text{lcm } \mathbb{Z}$.